

SENSITIVITY OF AN AUDIO-DEEFAKE LEADERBOARD TO THE RESAMPLING UNIT AND TO SOURCE, SPEAKER, AND ATTACK WEIGHTING

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ABSTRACT

We examine whether pairwise equal-error-rate (EER) comparisons on fixed detector scores remain separated from zero when resampling individual trials or jointly resampling speakers and attacks. Separation uses procedure-sensitivity bands adjusted over all system pairs. On ASVspoof 2021 DF, 26 of 28 pairs among eight systems are separated under trial resampling, compared with 18 under speaker-and-attack resampling; separation changes from 5/6 to 0/6 among organizer baselines and from 5/6 to 2/6 among self-supervised (SSL) detectors. An analogous SpoofCeleb comparison, using unstratified trial resampling on a different four-detector roster, falls from 6/6 to 3/6, so between-source mixture changes are not necessary. Alternative weights reverse five of six organizer-baseline orderings and one of six within-SSL orderings; the latter pair is already unseparated under trial resampling. All sixteen comparisons between the self-supervised detectors and the organizer baselines remain separated under both resampling methods with all twelve weighting rules, and under separate leave-one-source-out jackknife refits, whereas within-group separation changes with the resampling method and with source deletion. These are fixed-score sensitivity bands, not population confidence intervals.

Index Terms— Anti-spoofing, audio deepfake detection, evaluation methodology, clustered bootstrap, ranking uncertainty

1. INTRODUCTION

In 2021 the ASVspoof organizers tested every pair of submitted systems for significance, applied a Holm–Bonferroni correction, and published the deepfake task’s systems in four groups, noting that the top two were not separated [1]. We found no uncertainty analysis in the later journal overview [2], ASVspoof 5 [3], or Speech DF Arena report [4]; a 2026 self-supervised-model benchmark also reports point EERs [5]. AT-ADD reports trial-bootstrap rank stability [6]. Here we compare matched trial-versus-speaker/attack resampling of paired EER contrasts: XLSR-Mamba versus XLSR-Conformer loses separation under PW resampling (Fig. 1).

That matters more than it would if the analyses the field did run had represented the dependence. ASVspoof 2017 reported 95% EER confidence intervals and read their overlap [7], and 2021 tested every pair [1]; both use [8]. The three later benchmark reports named above contain no such analysis, although bootstrap EER spreads [9], Holm-corrected pairwise EER tests [10], and bootstrap intervals at a fixed threshold [11] do appear in recent detector and benchmark papers. Both variants of the cited test [8] treat trials within a class as independent Bernoulli draws; the dependent form accounts for two systems sharing trials, not for trials sharing speakers and attacks.

ASVspoof 2021 DF contains 533,928 evaluation trials but only 93 speakers and 110 attacks, and per-attack EERs for SSL-AASIST span 0.4–10.7%. Across the eight detectors, the one-way random-intercept ICC $\sigma_s^2/(\sigma_s^2 + \sigma_e^2)$ on bona-fide scores ranges from 0.15 to 0.77, where σ_s^2 and σ_e^2 are the speaker-intercept and residual variances. ICC alone does not establish dependence in paired EER differences; below we measure how paired contrasts respond to two declared perturbation laws on the joint scores, and repeat the comparison on two external datasets.

We audit two evaluation choices on fixed released scores: whether trials or speakers and attacks are resampled, and how observed source, speaker and attack groups are weighted. The primary question is whether pairwise separation survives those choices; we propose no new bootstrap and claim no identified population sampling law. On 21DF the six organizer-baseline pairs change from five separated to none, and on single-source SpoofCeleb the corresponding four-detector comparison changes from six to three. Alternative positive weighting rules then localize the stable conclusion: separation between the SSL-detector and organizer-baseline cohorts survives, whereas within-cohort ordering does not.

Scope. The primary set comprises eight score files released by the corresponding model authors or ASVspoof organizers, not the universe of public re-scores. We analyze all eleven Speech DF Arena score files for the same trials [4] separately. Four VCC spoof strata contain only 4/4/4/6 speakers, so no 21DF separation result transfers to new speakers or attacks. The bands and reweighting results describe fixed scores under declared procedures, not population confidence or safety.

2. RELATED WORK

This benchmark’s own reporting. ASVspoof 2021 Fig. 4(c) pairs systems with the Bengio–Mariéthoz test [8] under Holm correction and publishes group structure [1]; Wang and Yamagishi [12] showed that changing only the training seed moves ADD rankings—a variance component our intervals do *not* include.

Clustered resampling. Poh, Martin and Bengio [13] showed that biometric uncertainty is governed by users rather than sample count; Wu et al. carried user-aware resampling into speech [14] and developed a paired correlation-corrected two-system test [15]. Our trials instead have one speaker and, for spoof, one attack: two crossed factors [16], a delicate design [17, 18] distinct from the nested-observation designs studied in [19]. Our contribution is the audited fixed-data result: after EER threshold refitting and all-pair multiplicity are aligned, outputs change with the declared perturbation unit. We claim neither cell-level agreement with Fig. 4(c) of the

ASVspoof 2021 report nor population confidence.

Evaluation-design audits. In audio deepfakes, aggregation weights change conclusions [20], channel presentation can dominate model choice [21], and other audits question representations [22], attribution under controlled generator and speaker perturbations [23] and corpus overlap [24]. Speaker-conditioned detector-score sensitivity has also been probed through identity perturbations [25] and speaker-adversarial training [26], and TwinShift separately evaluates speaker and synthesizer familiarity [27]; here we vary resampling and evaluation weights on fixed scores. Simultaneous rank intervals are established [28]; adjacency is score-selected, so we control multiplicity over all 28 contrasts.

3. METHOD

3.1. EER contrast and separation event

Two deterministic systems on a fixed trial list have no sampling uncertainty until a perturbation law is declared. Scores are oriented so larger means bona fide and sorted in ascending order. At each ordered position, bona-fide mass at or below it is the false-reject rate (FRR) and spoof mass above it is the false-accept rate (FAR); the first position minimizing $|FRR - FAR|$ is selected and EER is their mean, without interpolation. Across the eight systems and twelve fixed weighting rules, evaluating EER only at distinct-score boundaries changes each point EER by less than 0.001 percentage point. For ordered pair (A, B) , $\Delta EER_{A,B} = 100(EER_A - EER_B)$ percentage points (negative favors A); every replicate refits both thresholds. A pair is *separated under procedure h* when its simultaneous band excludes zero; write this indicator as $I_{p,h}$. The identified object is the change in ΔEER and $I_{p,h}$ on the fixed scores, not population uncertainty.

3.2. Trial and speaker $\times$ attack bootstraps

The trial bootstrap independently resamples trials within class; for SpoofCeleb only, the archived plan samples the 91,130 trial indices from the pooled list with replacement, so class counts vary across replicates. In the speaker $\times$ attack product-weight (PW) bootstrap [18], speakers and spoof attacks are independently resampled with replacement; a spoof trial receives the product of its speaker and attack multiplicities, while a bona-fide trial receives only its speaker multiplicity. Speaker and attack labels define observed group-composition changes that trial resampling largely averages away. Product resampling probes joint sensitivity to those changes and may be conservative [18]. Both procedures use the same scores, threshold rule, $B=5000$ shared draws and all-pair multiplicity correction.

3.3. All-pair multiplicity for procedure bands

For procedure h , let $s_{p,h} = \text{sd}_b(\Delta_{p,h}^{(b)})$. We set $q_{.95,h}$ to the empirical 95th percentile of $\max_{p=1}^{28} |\Delta_{p,h}^{(b)} - \bar{\Delta}_{p,h}|/s_{p,h}$ and report $\hat{\Delta}_p \pm q_{.95,h} s_{p,h}$ [29]. The quantile centers deviations on the bootstrap mean, whereas the bands center on the original estimate; band inclusion is not an empirical reversal probability. The maximum runs over all 28 contrasts, so each procedure yields one simultaneous band construction; no population error-control guarantee is asserted. Because a population sampling law is not identified for 21DF, we call these procedure-sensitivity bands, not confidence intervals.

Reconstructing the cited test. Fig. 4(c) of Yamagishi et al. [1] does not disclose which Bengio–Mariéthoz variant produced its matrix, so we make no cell-level attribution. Both reconstructions of Fig. 4(c) (independent-trial and shared-trial) separate the same five of six baseline pairs; we claim no cell-level agreement. The published 34-system matrix has $\binom{34}{2} = 561$ tests; the largest p among our five reconstructed results, 9.7×10^{-6} , is below even its Bonferroni threshold $.05/561 \approx 8.9 \times 10^{-5}$, while the sixth has $p = .217$. The cited derivation assumes thresholds fixed outside the test set, so these are reconstructions of its adaptation to EER rather than exact finite-sample tests.

3.4. Alternative fixed weighting rules

Within each class, every named hierarchy level receives equal conditional mass and trials divide their terminal-group mass equally. Bona-fide rules are empirical trial weights, equal mass over the three sources (ASVspoof, VCC2018, VCC2020), and equal source $\rightarrow$ speaker mass. Spoof rules are empirical trial weights, equal mass over the five spoof strata (ASVspoof, VCC2018 HUB, VCC2018 SPO, VCC2020 Task 1, VCC2020 Task 2), equal mass over observed speaker $\times$ attack cells within stratum, and equal mass successively over stratum, attack family (the attack-type field of the official 21DF trial metadata), attack and speaker. Their 3×4 crossing gives 12 positive, class-normalized rules fixed without inspecting scores. For endpoint w^1 , the path from empirical weights is $w(\lambda) = (1 - \lambda)w^0 + \lambda w^1$, $0 \leq \lambda \leq 1$. We measure its distance by maximum classwise total variation, $\max_c (1/2) \sum_i |w_{c,i} - w_{c,i}^0|$.

4. EXPERIMENTS

Setup. The primary set contains four author-released self-supervised systems (XLSR-Mamba [30], XLS-R+SLS [31], XLSR-Conformer [32], SSL-AASIST [33]) and four organizer-released baselines (RawNet2, LFCC-LCNN, LFCC-GMM, CQCC-GMM). We use all 533,928 evaluation-phase trials for which every score is available. Fig. 1 shows all within-SSL pairs; Table 1 gives selected bands and group counts. The matched trial/PW comparison uses 5,000 replicates and seed 2026081604; seeds of the other analyses accompany their released outputs.

Primary 21DF result (Fig. 1). The matched all-28-pair bootstrap refits the same weighted EER in every replicate. Trial resampling separates 26/28 pairs and 5/6 organizer-baseline pairs; the PW bootstrap separates 18/28 and 0/6. A constrained PW bootstrap (1,000 draws, seed 20260824) resamples speakers and attacks within each bona-fide source or spoof stratum, redraws a replicate only when a class loses all support in a stratum (7 of 1,007 attempts, all VCC2018 spoof), and then rescales each source and stratum mass to its observed value; it also separates 18/28 and 0/6. This arm was designed after the main result and is a robustness check, not prospective evidence. Thus neither threshold handling nor movement of these eight class-specific masses is necessary for the contrast.

Single-source SpoofCeleb check. The pre-access chronology below is documented in the authors' records, without independent timing attestation. The analysis plan, archived on 29 August 2026 while official access was pending and released with its hash in the repository of Sec. 5, fixed the four detectors, score orientation, bootstrap laws, seed and six-pair endpoint before the first successful authenticated download. On all 91,130 evaluation trials [34], pooled trial resampling separates 6/6 pairs (a class-stratified recomputation

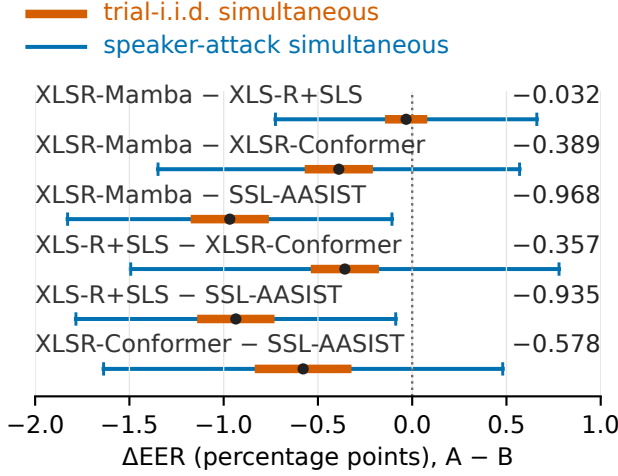


Fig. 1. All six within-SSL pairs on primary 21DF. Dots and right-hand labels are point gaps ($A - B$; negative favors A). Thick orange: trial bands; thin blue with end caps: PW bands. Both retain the full 28-pair simultaneous max- t construction. Five trial bands and two PW bands exclude zero; the point estimates do not change.

also separates 6/6) and the PW bootstrap separates 3/6. AASIST, SLS, SSL-AASIST and XLSR-Mamba have EERs 57.93%, 24.51%, 26.72% and 27.58%; this is an off-domain sensitivity check, not a comparison of competitive SpoofCeleb systems. Every base item has one bona-fide and nine spoof renditions under one source label, so between-source mass cannot move; speaker and spoof-attack multiplicities still vary under PW draws. After observing this result, separate speaker-only and attack-only bootstraps separated 5/6 and 3/6; attack-only matched the PW indicator vector. The archived plan did not bind the scoring code, so a later re-scoring from freshly cloned, commit-pinned detector repositories is reproducibility evidence only: three score files were byte-identical, the fourth differed by at most 1.44×10^{-6} , and every EER and separation indicator was unchanged.

Reading Fig. 1. Mamba-Conformer stays at -0.389 points but only its trial band excludes zero. Report the ordering as procedure-sensitive within the full 28-pair family, alongside the gap and both bands (Table 1). This neither estimates a reversal probability nor proves the trial conclusion false; it does not establish equality or determine a deployment ranking.

What 21DF supports. The 16 cross-cohort gaps are 19.5–23.7 points (supplement, Sec. S4) and remain separated after each leave-one-source-out refit. The two within-SSL pairs separated by the PW bootstrap (gaps 0.968 and 0.935 points) each cease to separate under one source deletion in the jackknife construction below. PW bands are 4.13 – $11.36\times$ wider than matched trial bands over all pairs (median $8.14\times$).

Alternative weights. Under at least one of the 12 fixed rules, six of the 12 within-cohort pairs reverse their point ordering (five within-baseline; the SSL one is XLSR-Mamba versus XLS-R+SLS, gap 0.032 points) and none of the 16 cross-cohort pairs does; one of the 11 nonempirical linear paths reverses before maximum classwise total variation .10. Recomputing both bootstraps with each rule’s weights as base weights (computed after the primary results, 5,000 draws per rule and bootstrap) separates all 16 cross-cohort pairs in all 24 rule $\times$ law cells; the closest band end-

Table 1. Top: selected paired EER differences and simultaneous bands (points); primary 21DF uses a 28-pair family, Arena a separate 55-pair family on the same trials. Bottom: primary pairs whose bands exclude zero, or whose ordering reverses under at least one of the 12 weighting rules.

XLSR-Mamba – XLSR-Conformer: -0.389			
Trial $[-0.570, -0.208]$		PW $[-1.348, 0.570]$	
Arena HuBERT-ECAPA – WavLM-ECAPA: -2.153			
Trial $[-2.761, -1.544]$		PW $[-5.241, 0.936]$	
Pair group	Trial	PW	Reversed
Within SSL	5/6	2/6	1/6
Within baselines	5/6	0/6	5/6
SSL vs. baselines	16/16	16/16	0/16

point to zero is -12.79 points. An exploratory post-result search found a witness reversing XLSR-Mamba minus XLS-R+SLS under the original EER sweep at maximum classwise total variation .010218, with bona-fide masses (.544, .244, .212) and spoof-stratum masses (.133, .234, .117, .223, .294), in the order above and rounded (greater precision in S5). The reversing sign also holds at distinct-score boundaries (supplement, Sec. S5). This is a found upper bound, not a minimum, and is too small to imply practical superiority. The search found no cross-cohort reversal, which is not a proof that none exists.

Alternative released scores on 21DF. On the eleven complete Speech DF Arena score files for the same 533,928 trials, trial and PW bootstraps separate 52/55 and 38/55 pairs; the median ratio of PW to trial bootstrap standard deviations over the 55 pairs is 8.45. For HuBERT-ECAPA versus WavLM-ECAPA (-2.153 points), the bands are $[-2.761, -1.544]$ and $[-5.241, 0.936]$. These are descriptive results for that separate score collection, not population confidence intervals.

Source deletions. The bona-fide sources ASVspoof, VCC2018 and VCC2020 contain 67, 12 and 14 speakers. Source deletion uses a third construction: with delete-one speaker, attack and observed-cell jackknife variances V_s , V_a and V_{sa} of each contrast, the interval is $\hat{\Delta} \pm q\sqrt{\max(V_s + V_a - V_{sa}, V_s, V_a, 0)}$, where q is a Gaussian max- t critical value recomputed over all 28 contrasts for each remaining subset (200,000 draws). On the full data this construction separates 18/28 pairs (16 cross-cohort, 2 within-SSL, 0 within-baseline), the same verdicts as the PW bootstrap. Leave-one-source-out refits preserve separation for all 16 cross-cohort pairs, remove both separated within-SSL pairs and separate three within-baseline pairs in at least one deletion. Thus cross-cohort separation survives these source changes, while some within-cohort separation decisions change with them.

ASVspoof 5 check. For SSL-AASIST, AASIST, XLS-R+SLS and XLSR-Mamba on all 680,774 Track 1 trials [3], resampling the 367 target speakers, the 370 bona-fide-only speakers and the 16 attacks as three independent multinomial draws changes the separated count from 6/6 under trial resampling to 5/6 and gives a median simultaneous-band width ratio of $27.27\times$; both criteria (at least one lost separation, median ratio at least 2) were fixed in the released plan before scoring completed. This check is intentionally EER-only because it transfers the paper’s Δ EER comparison; ASVspoof 5’s primary Track 1 metric is minDCF. It is not a system-performance replication or population claim.

Coverage boundary. The candidates are the raw exact-cell

inclusion–exclusion interval $\hat{\Delta} \pm 1.96\sqrt{V_S + V_A - V_{Sa}}$ with the jackknife variances defined above, its variance-floor version using $\max(V_S + V_A - V_{Sa}, V_S, V_A)$, and the 2.5–97.5 percentile PW band. We simulated 48 bivariate random-effects models on the 21DF trial-to-group structure: two regimes fitted to the RawNet2/LFCC-LCNN and XLSR-Mamba/XLS-R+SLS pairs, four interaction shares (0, .25, .50, .75: the fraction of the fitted spoof speaker-plus-attack variance moved into a speaker $\times$ attack interaction, total held fixed), Gaussian or t_5 effects, and target Δ of 0, 0.5 and 2 points, 1,000 replicates per cell (parameters in the supplement). Over the 24 low-EER cells the minimum nominal-95% coverages are .845, .845 and .855; all three are below .90 in the same 11/24 cells. This simulation evaluates pointwise candidate intervals under those imposed models, not the all-pair bands above; it does not establish that either model describes 21DF.

5. DISCUSSION

Reporting and limits. The complete 28-row band table, trial-to-source/stratum map, weighting formulas, 48-cell simulation parameters, the SpoofCeleb and ASVspoof 5 analysis plans with their hashes, and all artifact identifiers are in paper/SUPPLEMENT.md at github.com/rvirgilli/asvspoof2021-df-clustered-uncertainty, tag `icassp2027-submission`. The ASVspoof 5 SSL-AASIST and AASIST score files lack their originating run logs. Benchmark reports should publish trial-to-speaker/attack membership and source weights, not only trial counts, and report separation under a declared perturbation law and weighting next to the point EER. A band containing zero means the procedure does not separate that pair; it does not establish equality. Neither the primary-score analysis nor the Arena-score analysis supplies population confidence intervals or confidence sets for ranks.

6. CONCLUSION

On the eleven-system Arena scores for the same 21DF trials, separation changes from 52/55 under trial resampling to 38/55 under PW resampling. For the fixed 21DF score files, matched bootstraps change the separated organizer-baseline count from 5/6 to 0/6, and the single-source SpoofCeleb check, on a different four-detector roster, changes it from 6/6 to 3/6. Because SpoofCeleb has one source, between-source movement cannot explain the latter contrast; an attack-only bootstrap computed after the result matches its 3/6 indicator vector. All 16 pairs between the four SSL detectors and the four organizer baselines remain separated under both bootstraps, each leave-one-source-out refit of the declared jackknife construction and all 12 fixed weighting rules, and none reverses its point ordering in the searched directions; within-cohort separation changes with the resampling procedure and with source deletion, and within-cohort point ordering reverses under some weighting rules. Population confidence requires a defensible acquisition model and an inferential procedure valid under its dependence structure.

7. ACKNOWLEDGMENT

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8. COMPLIANCE WITH ETHICAL STANDARDS

This study analyzed previously collected ASVspoof 2021, ASVspoof 5 and SpoofCeleb recordings and detector scores. It involved no new recording or interaction with human participants; no ethical approval was required. The authors declare no competing interests.

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